

3) A resistive level inverter with $R_1 = 50k\Omega$ has following device parameters.

$$V_{TN}(V_{SB}=0) = 1.0V \quad \gamma = 0.5V^{1/2} \quad t_{ox} = 0.05\mu m \quad \mu_n = 500cm^2/Vs \quad W = 10\mu m$$

$$\text{and } L = 1.0\mu m$$

A ring oscillator is formed by connecting five identical inverters in closed loop. We are interested in finding the resulting oscillation frequency.

a) Find the delay times of the inverter for an ideal step input whose voltage swing is between V_{OL} and V_{OH} . T_{PHL} and T_{PLH} . It should be noted that the loading capacitance and gate capacitance of the following stage. For simplicity.

b) The rise and fall time are defined between 10% and 90% of the full voltage swing but for simplicity we will assume that $T_{fall} = 2 \cdot T_{PHL}$ and $T_{rise} = 2 \cdot T_{PLH}$.

Estimate the actual propagation delay T_{PHL} and T_{PLH} by using the rise and fall times of the inverter and the delay found in

c) Find the oscillation frequency from information in part (b)

Sol:
Given data:

$$V_{TN}(V_{SB}=0) = 1.0V \quad \mu_n = 500cm^2/Vs$$

$$\gamma = 0.5V^{1/2}$$

$$t_{ox} = 0.05\mu m$$

$$W = 10\mu m$$

$$L = 1\mu m$$

9 identical inverters $\rightarrow$ ring oscillator

assume $V_{DD} = 5V$

load capacitance = gate capacitance.

a) Ideal propagation delay T_{PHL} and T_{PLH} :

Gate capacitance

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} \approx \frac{3.45 \times 10^{-13}}{0.05 \times 10^{-6}} = 6.9 \times 10^{-9} F/m^2$$

$$C_{load} = C_g = C_{ox}WL$$

$$C_{load} = 6.9 \times 10^{-9} \times (10 \times 10^{-6}) (1 \times 10^{-6}) = 6.9 \times 10^{-19} F$$

Nmos Current

$$k_n = \mu_{n\text{ox}} \frac{W}{L} = (0.05) 6.9 \times 10^{-7} (10)$$

$$k_n = 3.45 \times 10^{-7} \text{ A/V}^2$$

Delay expansion

High-low (Nmos ON)

$$T_{PHL} \approx \frac{C_{load} V_{DD}}{I_n}$$

$$I_n = \frac{1}{2} k_n (V_{DD} - V_t)^2 = \frac{1}{2} (3.45 \times 10^{-7}) (4)^2 = 2.76 \times 10^{-6} \text{ A}$$

$$T_{PHL} = \frac{(6.9 \times 10^{-18}) (5)}{2.76 \times 10^{-6}}$$

$$\boxed{T_{PHL} = 12.5 \text{ ps}}$$

Low-High:

$$\begin{aligned} T_{PHL} &\approx 0.69 R_L C_{load} \\ &= 0.69 (50 \times 10^3) (6.9 \times 10^{-18}) \end{aligned}$$

$$\boxed{T_{PHL} = 238 \text{ ps}}$$

b) Actual propagation Delay $\Rightarrow$

$$T_{fall} = 2 t_{PHL}, \quad t_{rise} = 2 t_{PLH}$$

$$T_{fall} = 25 \text{ ps}, \quad T_{rise} = 476 \text{ ps}$$

$$T_{PHL} = \frac{T_{fall}}{2} = 12.5 \text{ ps}$$

$$T_{PLH} = \frac{T_{rise}}{2} = 238 \text{ ps}$$

c) Oscillation frequency

$$f = \frac{1}{2N(t_{PHL} + T_{PLH})} = \frac{1}{4.5 \times 10^{-9}}$$

$$\boxed{f = 222 \text{ MHz}}$$